

Peter Zhang

pzhang04@stanford.edu | (408) 771-8845 | [Personal Website](#)

Education

STANFORD UNIVERSITY

BS in Computer Science and Mathematics

Stanford, CA

May 2027

Experience

Stanford University & Kobilka Lab (Nobel Laureate Brian Kobilka)

Stanford, CA

Primary Researcher (Project Lead)

June 2025 – Now

- Architected a deep reinforcement learning framework for generative molecular optimization, discovering 5 novel lead candidates (U.S. patents filed).
- Achieved a predicted 28,000-fold selectivity for the CB2 receptor, surpassing the widely recognized in vitro benchmark by nearly 30x.
- Built a large-scale multi-objective screening pipeline over 1.5M+ compounds using reward-guided generative search and automated docking evaluation.

Stanford Vision & Learning Lab with Professor Ron Fedkiw

Stanford, CA

Researcher (LLM & Diff. Modeling)

September 2024 - Now

- Developed differentiable neural rendering systems combining transformer-based feature encoding and volumetric priors for robust 3D hair reconstruction.
- Built distributed PyTorch GPU training pipelines with scalable augmentation and fine-tuning workflows.
- Applied transformer-based neural rendering methods to improve robustness under occlusion and pose variation.

Insilico Medicine (HK:3696)

Shanghai, China

AI Research Intern

July -August 2024

- Engineered AI agents to automate high-volume, repetitive workflows within the drug discovery pipeline, reducing manual intervention by 30 hours/week.
- Designed agentic pipelines integrating large language models and biological tooling, reducing repetitive research operations by 30+ hours/week.
- Collaborated with cross-functional teams of biologists and data scientists to translate complex pharmaceutical requirements into scalable machine learning solutions.

Reachly-AI (Academia Research Helper)

Stanford, CA

Founder

November 2025 - Now

- Implemented LLM-based systems for profile analysis, powering an academic outreach platform for students.
- Built scalable backend workflows for document parsing, professor relevance scoring, and email generation, improving outreach efficiency by saving 20+ hours per week.

Leadership and Activity

- **Immerse the Bay (XR Hackathon)** Grand Prize + UltraLeap + XR Trailblazer Winner (1 out of 300).
- **Sixth author**, *Realizing optimal radiative cooling walls in building-energy nexus via asymmetric emissivity*, Nexus (Cell Press)

Skills

- **Languages:** Python, C++, Java, Scheme | Native in both English and Mandarin Chinese.
- **Machine Learning:** PyTorch, Distributed Training, Transformer Models, Reinforcement Learning, Diffusion Models, Neural Rendering, LLM Fine-Tuning, Multimodal Learning
- **Infrastructure:** CUDA, GPU Training Pipelines, Docker, Linux, Git, High-Performance Computing (SLURM)
- **Technical focus:** LLM Fine-Tuning, Neural Rendering, Multimodal ML, Data Pipeline Design.